

(Non-Newtonian Viscosity)

Pre-Lab Plan

Unit Operations Lab 2

9/24/2019

Group 1, Tuesday, AM section

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1. Executive Summary:

The main objective of this experiment was to prepare non-Newtonian fluid using Carbopol in order to determine its viscosity parameters. These parameters were determined using the following formula: $m \cdot \left[\frac{3n+1}{n}\right]^n \cdot \left(\frac{Q}{\pi R^3}\right)^n = \left(\frac{\Delta P}{2L}\right) R$ which is used to calculate the viscosity of the non-Newtonian fluid. The formula is known as the Ostwald-deWaele viscosity model which was used to determine the parameters “m” and “n” using the obtained data during the experiment. 0.8 wt% solution of Carbopol was used in order to prepare the non-Newtonian fluid by mixing the Carbopol with warm water. In conclusion, based on the results obtained throughout the experiment, a non-Newtonian fluid, the flow rate decreases as the length of the capillary tube increase and the diameter of the tube decreases.

2. Introduction

Viscosity is a term used in Chemical Engineering used when describing the physical property of how the fluid acts when transported or heated, usually having SI units of (Pa * s) . Furthermore, viscosity is consistent with the thickness of fluid. If the thickness of fluid is low, the viscosity of the fluid is also low and vice versa [1]. For example, water is considered to be a thin fluid, as a result, it has a low viscosity. Viscosity by definition is the ratio of the shear stress to the velocity gradient in the fluid normal to the shearing surface [1]. Fluids that following this relationship and depending on shear stress or shear rate are known as Non-Newtonian fluids [1]. An example of non-Newtonian fluids from everyday life is Ketchup, blood, toothpaste. The viscosity of these fluids decreases as the applied stress increases.

For this experiment, 0.8wt% aqueous solution of Carbopol 940 was prepared and used as the non-Newtonian fluid. The main objective of this experiment is to determine the parameters “m” and “n” from the Ostwald-deWaele viscosity model in order to establish the relationship between viscosity of non-Newtonian fluid and pressure, length, and diameter of the tube Carbopol is known as soluble polymer that is available as white powder that comes in different grades. For the purpose of this experiment, Carbopol 940 was used which turned to a thick gel as its pH was between 6.5-7.5 [2] for the Carbopol to act as a Non-Newtonian fluid. Moreover, experimental data is needed to calculate viscosity, An example would be to use a viscometer that uses a pressurized vessel and varying capillary tubing. For this experiment the HagenPoiseuille equation for Newtonian fluids, one may determine the fluid viscosity from measurements of flowrate and pressure drop, which follows the equation $Q = \Delta P \pi R^4 / 8 \mu L$ can not be used because our fluid is non-newtonian. Instead we used the the Non-newtonian equation as followed a power-law relationship (also known as the Ostwald-deWaele Model)[1].

$$m \left[\frac{3n + 1}{n} \right]^n \cdot \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} \right) \cdot R$$

Where:

ΔP : the pressure drop from the viscometer

R : the radius of the capillary tube

Q : the volumetric flow rate

L : the length of the capillary tube

n : the coefficient representing slope for the Ostwald-deWaele Model

m : a coefficient of the Ostwald-deWaele Model

To find the viscosity, the mass flow rate needed to be calculated by timing the mass of the carbopol rate and using the density (200 kg/m^3) to calculate the volumetric flow rate with pressure difference, with length and radius of the varying capillary tubes of the pressurized viscometer. The different variables used in the experiment was pressures of (10, 25, 35, and 45 psig), lengths of (3, 6, 9 in), and diameter of (0.062, 0.046 in) were used in this experiment. Then the values were inserted into the Ostwald-deWaele model, where m and n are the parameters then plot the $\log(\Delta P \cdot R^2 L / \eta)$ vs. $\log(Q \pi / R^3)$ will give a slope of ' n ' and an intercept value of $\log\{m[3n+1] n\}$. From the slope and intercept values obtained from the graph, the parameters m and n can then be calculated as a function of capillary diameter and length [1].

In the industry a company can use capillary viscometer if a fluid is needed to be tested to find its viscosity. Understanding how the fluid acts relating to the L/D and Pressure is important in the proper design of a plant. The only way to understand the fluids properties is through experimental data. Experimental data could be used to plot how the differences in pressure and capillary tubes affect the fluid to determine the viscosity, which can be used to determine appropriate piping dimensions.

3. Theory:

Newtonian fluids are fluids that have a low viscosity, which relates how a fluid's resistance to flow. For example of this would be a fluid like water, with usual properties like a constant viscosity that does not change. For these fluids, an increase in the shear rate causes a proportional increase in the velocity gradient [1]. Though the viscosity change their flow behaviour or viscosity only changes with changes in temperature or pressure [4]. Newtonian fluids are usually single and phase non polymer substances. As stated before, Newtonian fluids follow the HagenPoiseuille equation where the n coefficient is equal to 1.

Non-Newtonian fluids exhibit a relation between shear stress and the shear rate. In fact, non-Newtonian fluid have a different behaviours as stress is applied based on either the amount of stress applied or the time required to apply the stress. In fact, there are four different behaviors of non-Newtonian fluid, which are Thixotropic, Rheopectic, Shear thinning and shear thickening. Both Thixotropic and Rheopectic are behaviors that describe how viscosity decreases with the applied stress over time. An example of Thixotropic behaviour is honey. Unlike the Thixotropic, Rheopectic behavior shows how viscosity increases with the applied stress over time. Cream is an example of Rheopectic behaviors. In addition, shear thinning and shear thickening are behaviors that describe the correlation between the viscosity and the amount of applied stress. Sauces is an example of shear thinning behaviour which shows as the amount of applied stress

increases, the viscosity decreases. However, shear thickening shows the opposite which is as the amount of the applied stress decreases, the viscosity of fluid increases.

4. Results:

Carbopol is a non-newtonian fluid that presents behavior that can be modeled using a power law assumption. In this experiment, the Ostwald-deWaele Viscosity model was used. The factors affecting this model were the L/D ratio, pressure drop, and the flow rate. Through graphical analysis of these various values, it was possible to get the values of coefficients m and n.

L/D	3in/0.046 in	3in/.062 in	6in/0.046 in	6 in/0.062in	9 in/0.046 in	9 in/0.062 in
n	2.3907	2.2478	3.3405	7.4018	2.5886	4.1227
m	3.8463E-06	1.58742E-05	2.7556E-09	4.69725E-23	5.60E-08	1.94E-11

Table 1: Table of the values of M and N sorted by the size

In this Table 1 it is seen that 6in/0.062in capillary tube gives a great outlier. It was found throughout the experiment that the longer the capillary tube is and the smaller the diameter is, the more unreliable data became. This is because the flow rate would be greatly reduced giving smaller sample sizes more easily affected by experimental error. There was also the issue of bubbles formed during the creation of Carbopol which would more heavily impact the thinner and longer capillary tubes.

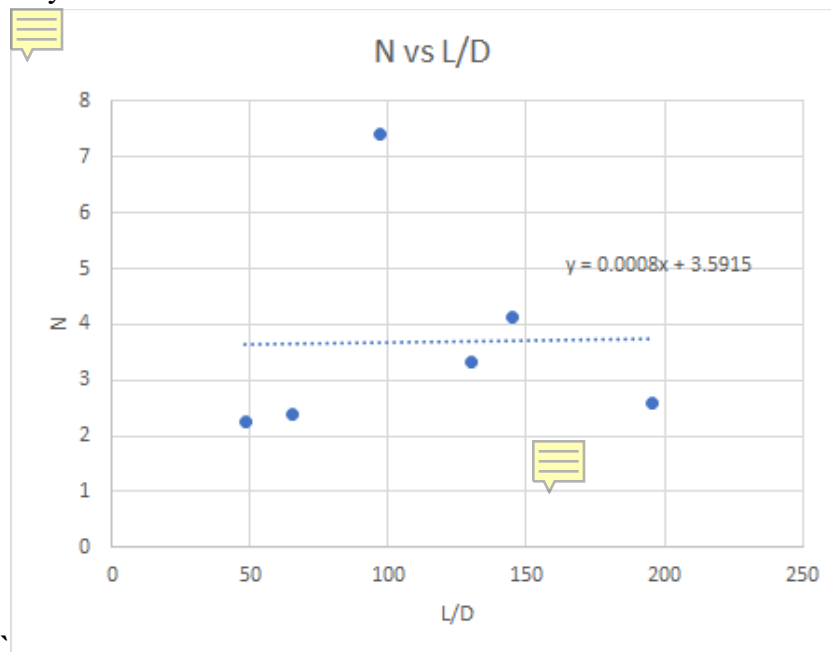


Figure 1: Figure of N values vs L/D ratio

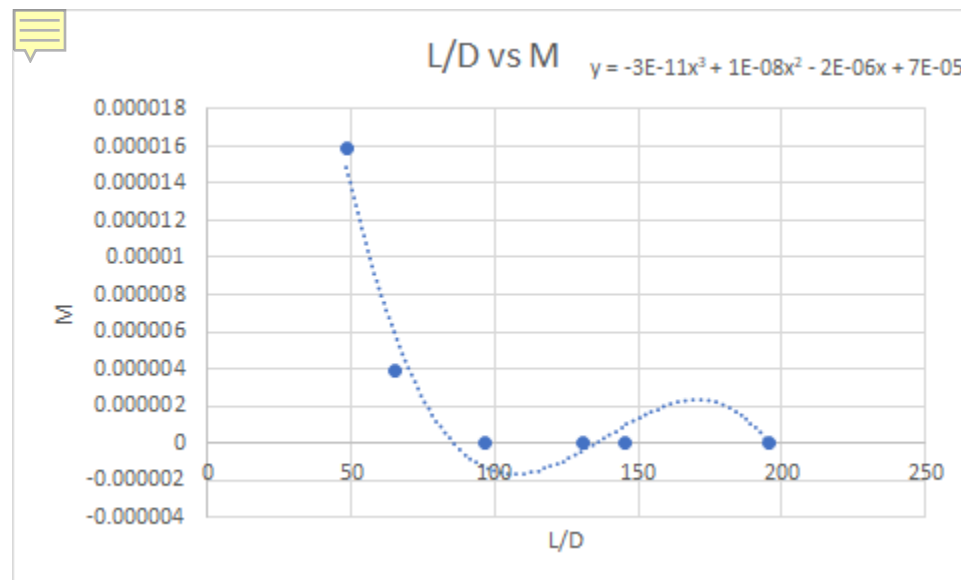


Figure 2: Figure of M values vs L/D ratio

In **Figure 1 and 2** the relationship of M vs L/D and N vs L/D is shown. It seems that N slightly increases with L/D while M decreases with L/D. However, these relationships are tenuous. The data is not very clear on what the relationship is.

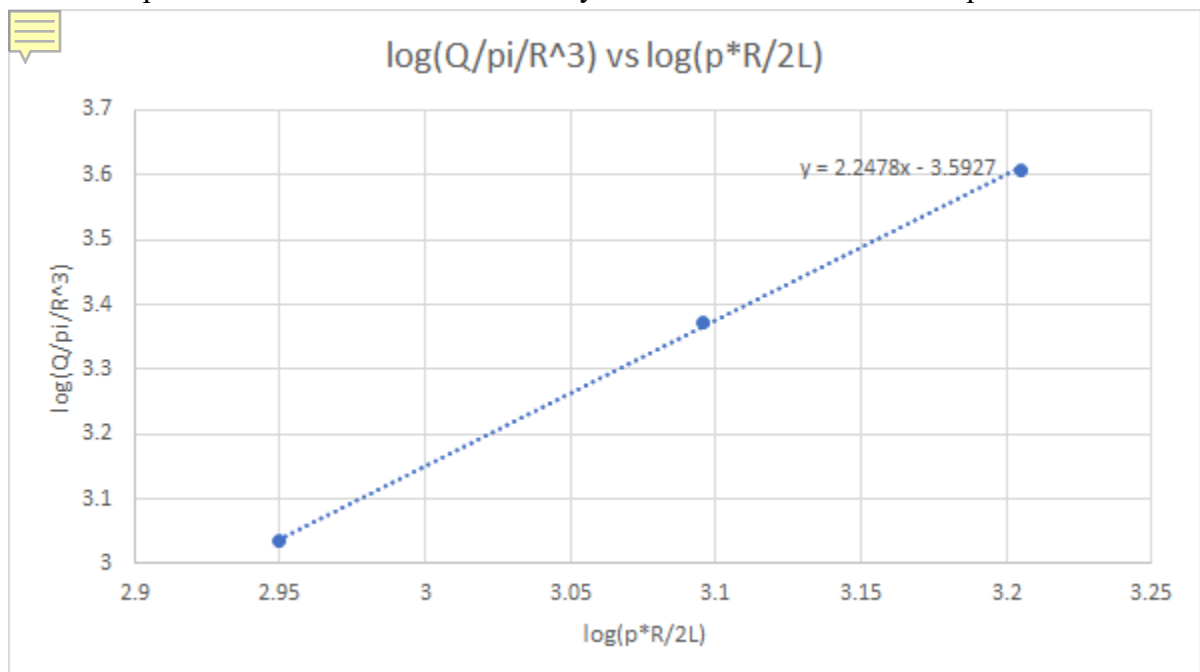


Figure 3: ostwald-deWaele model graph for length of 3 in and diameter of 0.062 in

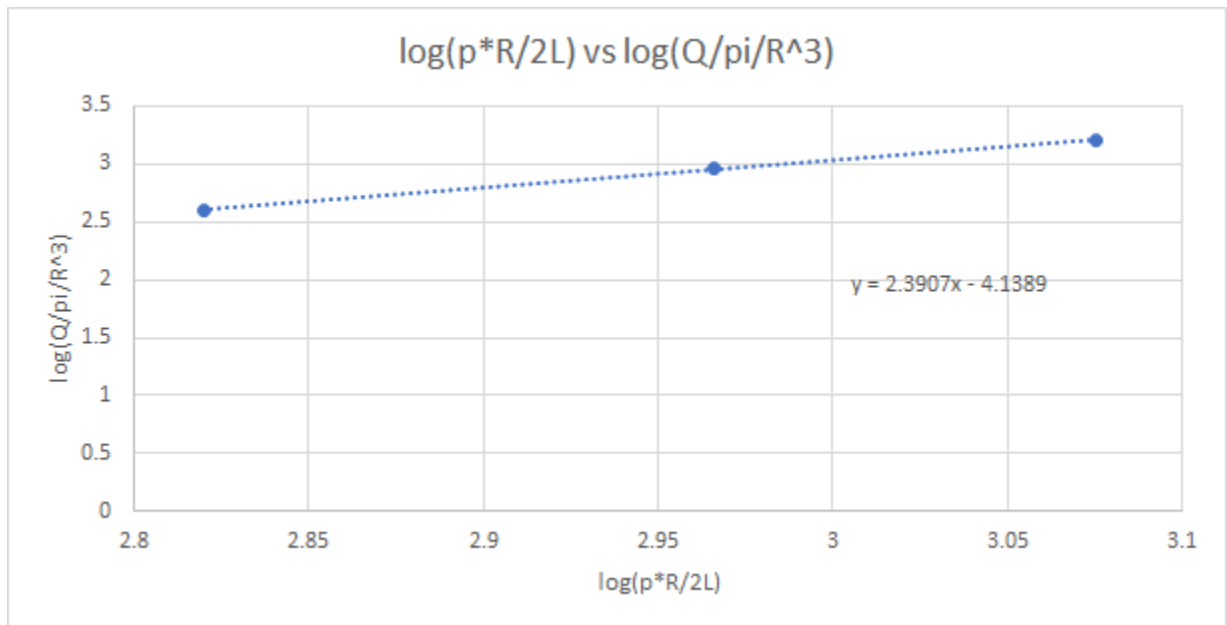


Figure 4: ostwald-deWaele model graph for length of 3 in and diameter of 0.046 in

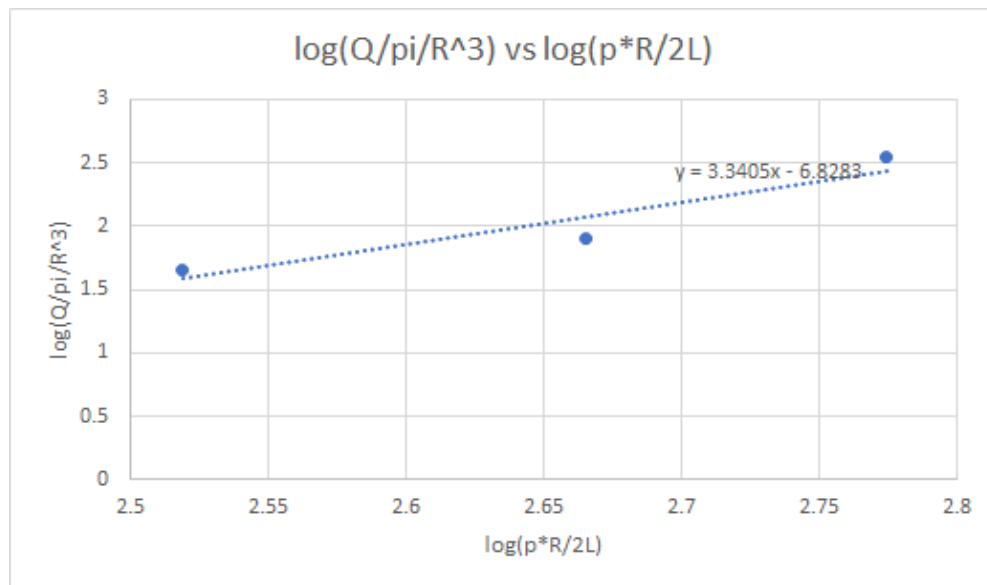


Figure 5: ostwald-deWaele model graph for length of 6 in and diameter of 0.046 in

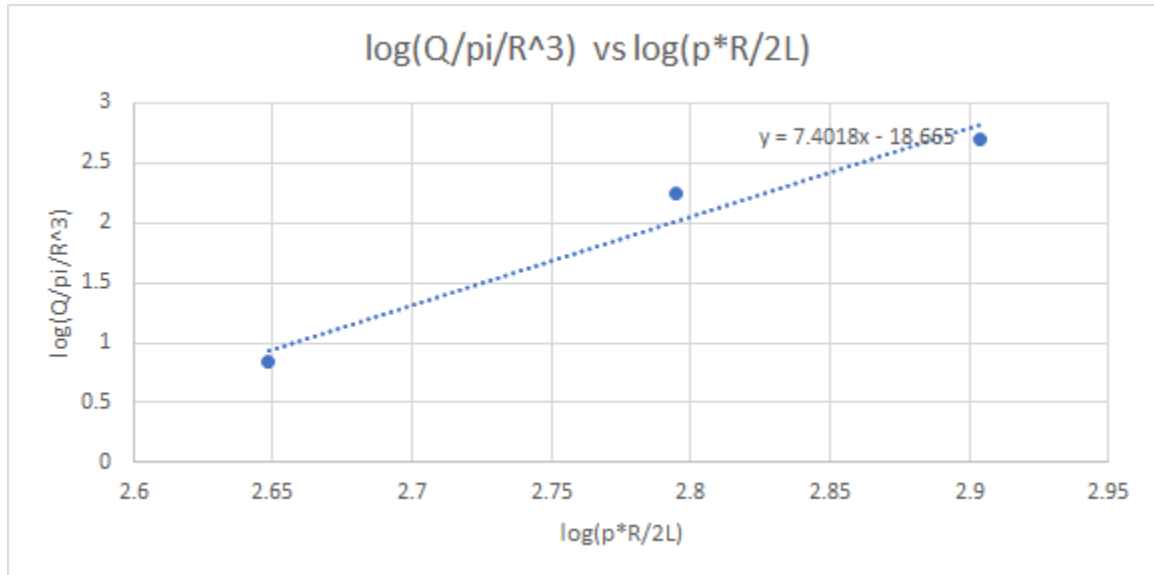


Figure 6: ostwald-deWaele model graph for length of 6 in and diameter of 0.046 in

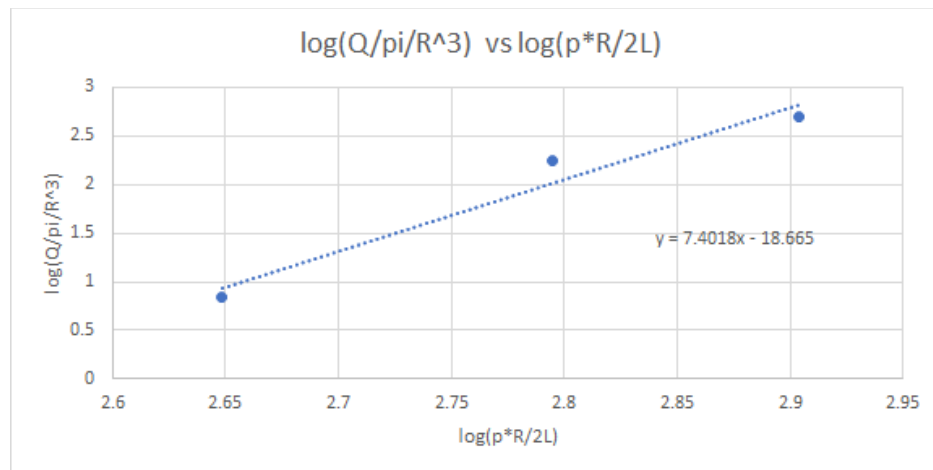


Figure 7: ostwald-deWaele model graph for length of 6 in and diameter of 0.062 in

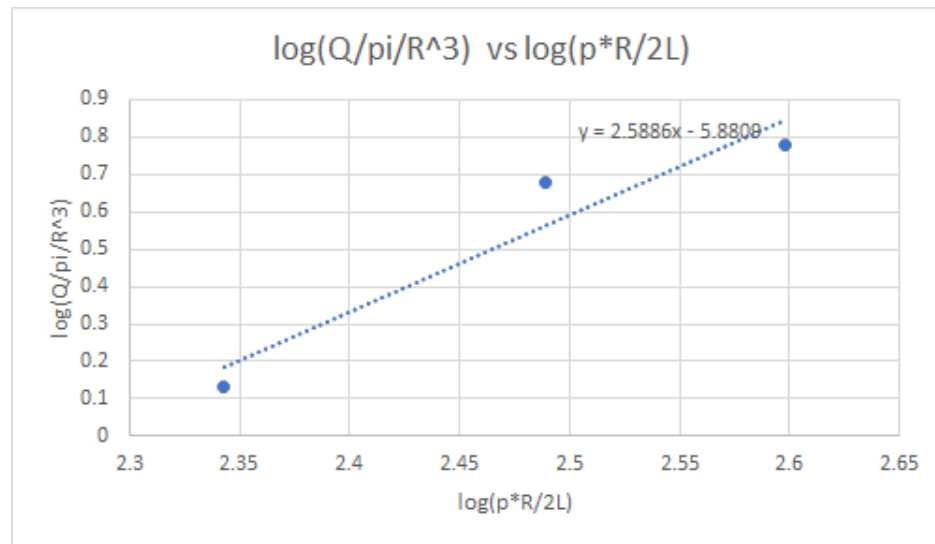


Figure 8: ostwald-deWaele model graph for length of 9 in and diameter of 0.046 in

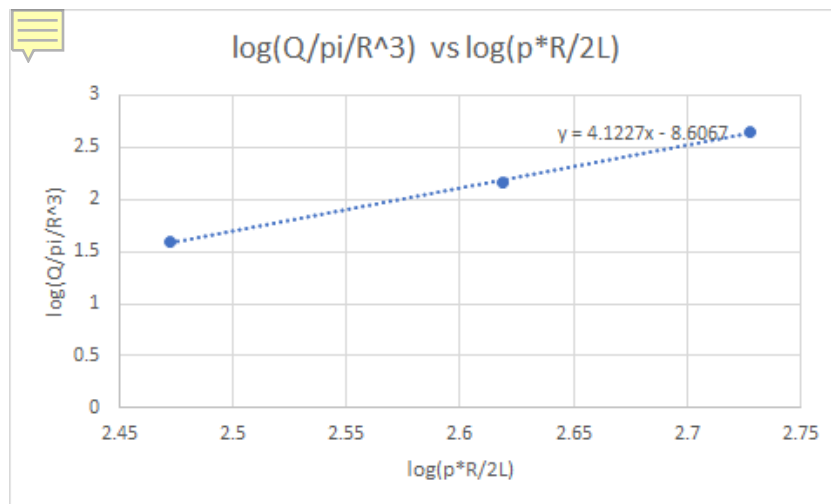


Figure 9: ostwald-deWaele model graph for length of 9 in and diameter of 0.062 in

In figure 3-9 the graphs to obtain the coefficients m and n are shown. There are a few points that deviate from trends but for the most part shows a strong linear relationship. These values of m and n would allow us to graph shear rate and shear stress. It is also important to note that although values were collected for some capillary tubes at 10 psi, these values were ultimately ignored since at such a low pressure the flow became too slow and noisy.

5. Discussion:

The relation found in through the calculations and graphs the “m” and “n” were related by length, radius, and volumetric flow rate. It was determined that the “m” coefficient to have the units of (Pa*s) the same units as viscosity because it follows a trend close to apparent viscosity and was seen to be related in literature, and “n” to have no units as it is treated as a factor. These two coefficients have different relations with L/D which are as L/D increases “m” increases and “n” decreases. However, this trend was not found in our experiment. The trend could be due to a variety of factors affecting the accuracy of our data. Another factor could also be that chosen pressures that are different as to see a larger contrast. Right now in the N vs L/D graph we see no relationship and in the M vs L/D graph we see something resembling almost an inverse relationship. There is no linearity that is expected.

6. References:

1. Zdunek, Alan. “ChE 382: Unit Operations Laboratory – Viscosity of Liquids.” 1-6. 2005. Print.
2. Lubrizol, “Neutralizing Carbopol® and Pemulen™ Polymers in Aqueous and Hydroalcoholic Systems” Technical data sheet, May 31, 2011. Available at: [Thickening-Properties.pdf](#). Accessed: 9/30/2019
3. Al-Malah, Kamal. “Rheological Properties of Carbomer Dispersions.” *University of Bahrain, Isa Town, Bahrain*, 2006.
4. “Non-Newtonian Fluids.” *Science Learning Hub*, April 12 2012, www.sciencelearn.org.nz/resources/1502-non-newtonian-fluids. Accessed: 9/30/2019

7. Appendix I: Data Tabulation/Graphs:

	length/diameter											
	3 in/0.046 in		3 in/0.062 in		6 in/0.046 in		6 in/0.062 in		9 in/0.046 in		9 in/0.062 in	
psig	secs	mass (g)	secs	mass (g)	secs	mass (g)	secs	mass (g)	secs	mass (g)	secs	mass (g)
10			120	126.62			300	120.9			391	120.3
25	120	126.06	120	160.2	240	121.49	120	126.81	587	120.21	200	122.68
35	90	130.51	60	163.54	240	122.49	120	138.65	300	120.29	140	126.58
45	79	136.01	30	157.62	180	127.93	60	138.66	320	120.35	120	136.76

Figure 10: Data collected during class

The mass collected during the experiment was calculated by subtracting the weight of the beaker at 120.26 g. Then the flow rate was calculated by dividing the mass by secs. With that, we had all values to graph.

	length/diameter 3 in/0.046 in	3 in/0.062 in	6 in/0.046 in	6 in/0.062 in	9 in/0.046 in	9 in/0.062 in
p.s.i	volumetric flow rate (m ³ /s.ecs)	volumetric flow rate (m ³ /s.ecs)	volumetric flow rate (m ³ /s.ecs)	volumetric flow rate (m ³ /s.ecs)	volumetric flow rate (m ³ /s.ecs)	volumetric flow rate (m ³ /s.ecs)
10		0.000000265		1.06667E-08		5.11509E-10
25	2.87917E-07	1.66417E-06	2.875E-08	2.72917E-07	8.51789E-10	6.05E-08
35	5.77778E-07	3.60667E-06	4.95833E-08	7.6625E-07	3E-09	2.25714E-07
45	1.06633E-06	6.22667E-06	2.17222E-07	1.53333E-06	3.75E-09	6.875E-07

Figure 11: Calculations of flow rate for the log vs log graphs

8. Appendix II: Error Analysis:

During the experiment, there are some areas where systematic errors occurred. For instance, the weight balance was unstable, as a result, the readings of the obtained fluid were off by ~ 0.3 g. whether or not the plastic container used to collect the obtained fluid was clean is another example of systematic error that slightly affected the accuracy of the weight reading. The same plastic container was used for all the trails, and as the plastic container was cleaned in each trail, a slight decrease in the weight reading was observed. In addition, timing is an example of random error happened during the experiment. A difference of ~ 0.5 s occurred as difference when starting/stopping the watch between each trail. As a result, the mass flow rate was slightly affected.

Furthermore, the uncertainty determined by calculating the standard deviation of each capillaries trials using the following equation:

$$S = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}}$$

Where:

S : the standard deviation

N: the number of measurements

(N-1): the degree of freedom

$(x_i - \bar{x})$: the difference between the experimental value and the average of the measurements.

For the purpose of this experiment, the standard deviation was calculated using built-up function in spreadsheet excel. To determine the standard deviation of parameter n, the following command was used in a cell in excel (=STDEV(#1, #2,#3, ...)) and then highlighted the calculated values of the parameter m, the same process was done to calculate the standard deviation of the parameter n. Therefore, the following results were obtained:

L/D	3in/0.046 in	3in/.062 in	6in/0.046 in	6in/0.062 in	9 in/0.046 in	9 in/0.062 in
n	2.3907	2.2478	3.3405	7.4018	2.5886	4.1227
m	3.8463E-06	1.58742E-05	2.7556E-09	4.69725E-23	5.60E-08	1.94E-11
std dev of m:	1.952826652					
std dev of n:	6.34955E-06					

Figure 12: Standard Deviation for error analysis

9. Appendix III: Sample Calculations:

➤ Determining the weight of Carbopol powder (g):

In order to prepare 0.8wt% Carbopol solution, the following calculation was performed to determine the mass (g) needed of Carbopol by assuming the following about of water is required.

$$1.5 \text{ L } H_2O * \left(\frac{1000 \text{ g water}}{1 \text{ L}} \right) * \left(\frac{0.8 \text{ g Carbopol}}{100 \text{ g water}} \right) = 12 \text{ g Carbopol}$$

➤ Determining the value of coefficient m and n:

In order to find the coefficient value m, the relationship $\log\left(\frac{Q}{\pi R^3}\right)$ vs $\log\left(P * \frac{R}{2L}\right)$ graphed. In this graph the slope is n and the y intercept is equal to $\log\left(m\left(\frac{3n+1}{n}\right)^{-n}\right)$. If the equation is rearranged and solved for m the result is:

$$m = 10^{y\text{-intercept}} * \frac{(3 * n + 1)^{-n}}{n}$$

For example when calculation the m value for a capillary tube with a length of 3 in and diameter of 0.046 in, the n value was found to be 2.3907. The y-intercept was -4.31889. This graph can be seen in figure 4. The resulting calculation was:

$$m = 10^{-4.31889} * \frac{(3*2.3907+1)^{-2.3907}}{2.3907} = 3.84E-06$$

➤ Determining the flow rates:

The results of this calculation are shown in **Figure 11**. In **Figure 10** the amount of Carbopol collected over a certain time was recorded. From the amount of time passed and the amount of Carbopol collected it is possible to calculate the volumetric flow rate. This is done by first subtracting the weight of the beaker from the weight of the amount collected, converting the mass to volume, and then dividing by the amount of time passed. Take for example the calculation at L/D of 3 in/0.046 in at 25 psi.

$$\begin{aligned} \text{mass of carbopol collected (g)} \\ &= \text{mass of beaker + carbopol} - \text{mass of beaker} \\ &= 126.06 - 120.11 = 5.95 \end{aligned}$$

$$\begin{aligned} \text{volume of carbopol collected (cm}^3\text{)} \\ &= \text{mass of carbopol collected / density of carbopol} = 5.95/0.2 \\ &= 29.75 \end{aligned}$$

$$\begin{aligned} \text{volume flow rate (m}^3\text{/s)} &= \text{volume of carbopol collected / time to flow/100}^3 \\ &= 2.47917E - 07 \end{aligned}$$

10. Appendix IV: Individual Team Contributions

NAME: Christian Sanchez

	TIME (HOURS)	<u>DESCRIPTION</u>
<u>Operator (Both Lab Days)</u>	6	
<u>Pre-Lab Editing</u>	1	
<u>Experimental Objective (Prelab)</u>	1	
<u>Apparatus (prelab)</u>	1	
<u>Materials & Supplies (prelab)</u>	1	
<u>Procedure (prelab)</u>	1	
<u>Project Safety Evaluation (prelab)</u>	1	
<u>Final Lab Editing</u>	1	
<u>Executive Summary (final lab)</u>	1	
<u>Introduction (Final lab)</u>	1	
<u>Theory (Final Lab)</u>	1	
<u>Results (final lab)</u>	1	

<u>Discussion (final lab)</u>	1	
<u>Conclusion (final lab)</u>	1	
<u>References (final and/or prelab)</u>	1	
<u>Data Tabulation/Graphs (final)</u>	1	
<u>Error Analysis (final lab)</u>	1	
<u>Sample Calculations (final Lab)</u>	1	
<u>Power Point Presentation</u>	2	
<u>Total</u>	25	

NAME: Advith Vashist

	<u>TIME (HOURS)</u>	<u>DESCRIPTION</u>
<u>Operator (Both Lab Days)</u>	6	
<u>Pre-Lab Editing</u>	1	
<u>Experimental Objective (Prelab)</u>	1	
<u>Apparatus (prelab)</u>	2	

<u>Materials & Supplies (prelab)</u>	1	
<u>Procedure (prelab)</u>	1	
<u>Project Safety Evaluation (prelab)</u>	2	
<u>Final Lab Editing</u>	2	
<u>Executive Summary (final lab)</u>	1	
<u>Introduction (Final lab)</u>	1	
<u>Theory (Final Lab)</u>	1	
<u>Results (final lab)</u>	1	
<u>Discussion (final lab)</u>	1	
<u>Conclusion (final lab)</u>	1	
<u>References (final and/or prelab)</u>	1	
<u>Data Tabulation/Graphs (final)</u>	1	
<u>Error Analysis (final lab)</u>	1	
<u>Sample Calculations (final Lab)</u>	1	
<u>Power Point Presentation</u>	1	

<u>Total</u>	<u>28</u>	
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NAME: Fatimah AL Ibrahim

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0.5	
Final Lab Editing	3	
Executive Summary (final lab)	2	

Introduction (Final lab)	2	
Theory (Final Lab)	1	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	1	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	2	
Sample Calculations (final Lab)	1	
Power Point Presentation	2	
Total	25.5	